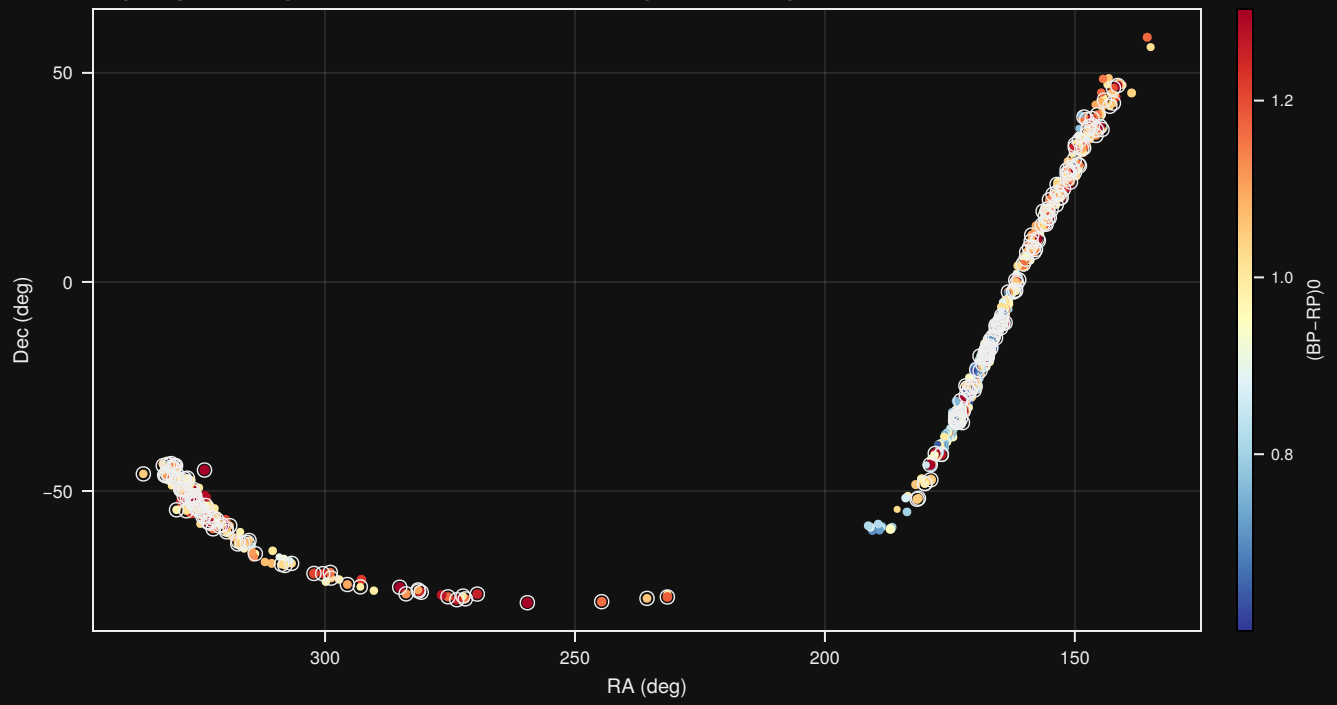
Orphan

MEMBERSHIP	
Stars in catalog	867
with Gaia GSP-Phot	515 (59%)
with radial velocity	247
Stream id (sID)	48
GEOMETRY · great-circle fit	
Centroid RA, Dec	173.1, -24.4 deg
Centroid l, b	275.4, 34.0 deg
Pole RA, Dec	252.5, 15.4 deg
Length (φ1, 2–98%)	179.0 deg
Width σ(φ2), robust	1.39 deg
KINEMATICS	
pmRA* med ± σ	-2.28 ± 1.27 mas/yr
pmDE med ± σ	1.58 ± 1.65 mas/yr
μφ1* med ± σ	-2.54 ± 1.20 mas/yr
μφ2 med ± σ	-1.68 ± 0.55 mas/yr
Parallax (cat.) med	0.026 ± 0.154 mas
Parallax (wtd, ZP-corr.)	0.047 ± 0.010 mas
Distance	d = 21.5 ± 4.8 kpc
Physical length	67.15 kpc
RV median ± σ(MAD)	163.9 ± 80.2 km/s
RV gradient dV/dφ1	-2.35 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	13.3–20.0 (18.1)
(BP−RP)0 median	0.93
GSP-Phot [M/H] (n=515)	-1.63 ± 0.87
GSP-Phot Teff median	5142 K
RUWE > 1.4	0.2 %
PM clusters (largest)	9 (709 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	867 | 867
n_RV catalog | paper	247 | 247
<[Fe/H]> paper	-1.90
Stellar mass (paper)	130000 Msun
Pole offset vs paper	4.3 deg
Remark	Massive system

Sky map · catalog members (n = 867) · size ∝ brightness · ring = has RV

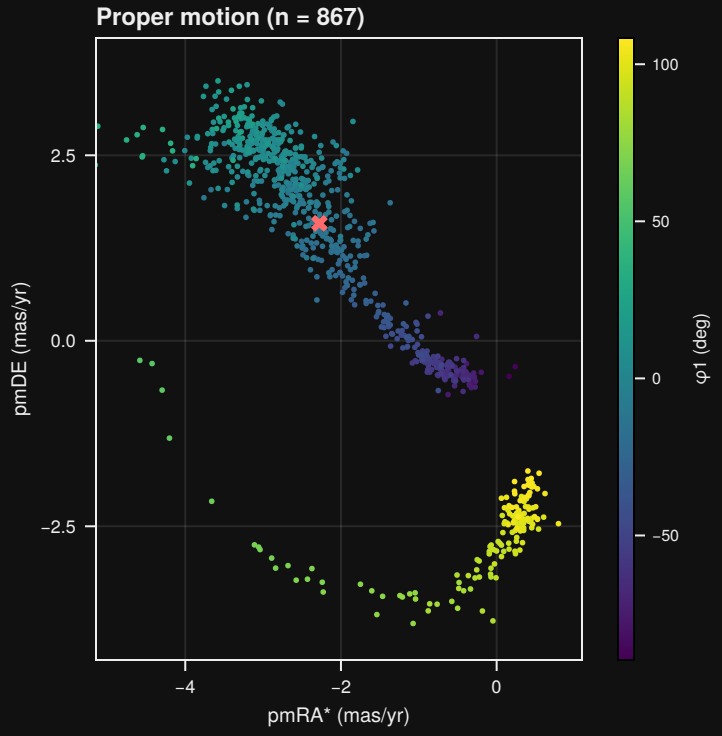


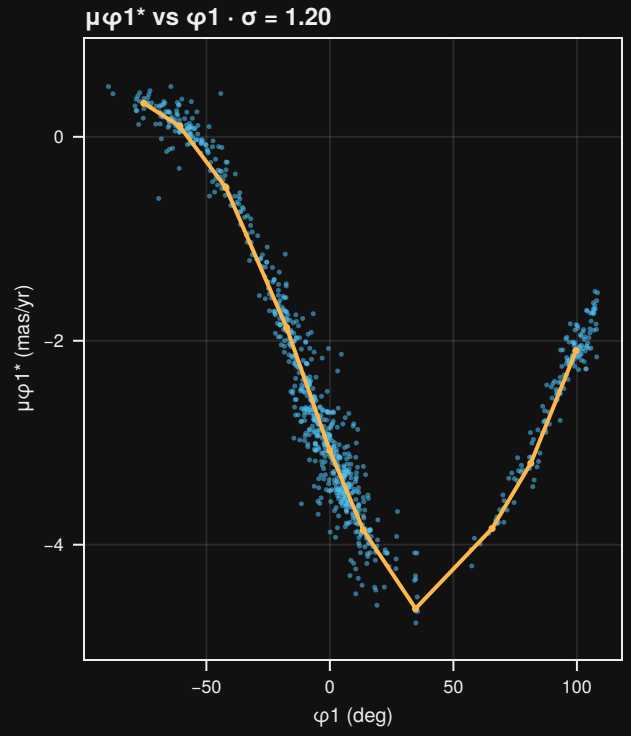
Stream frame (great-circle fit) · length 179.0° · width σ 1.39°

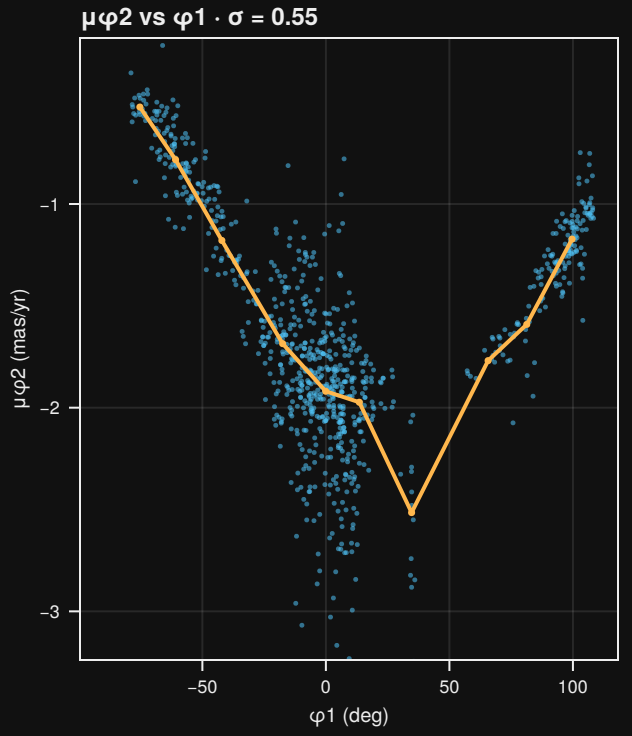


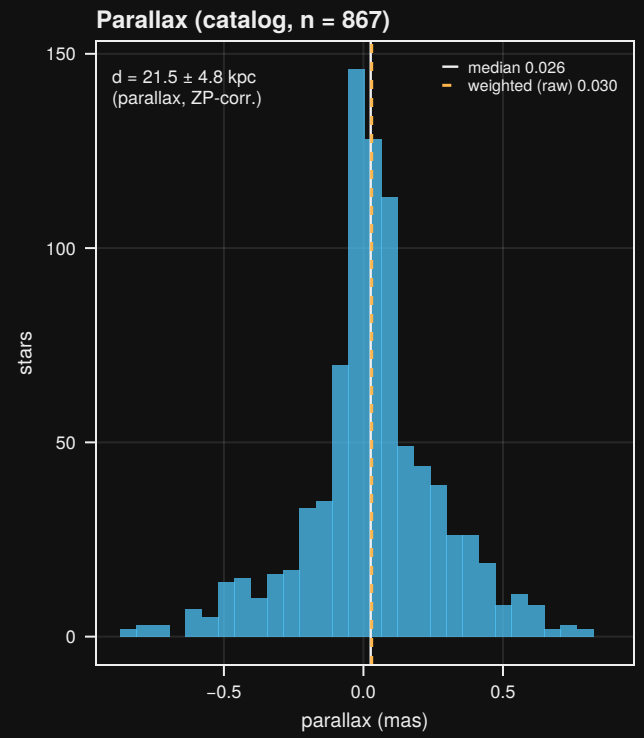
Orphan

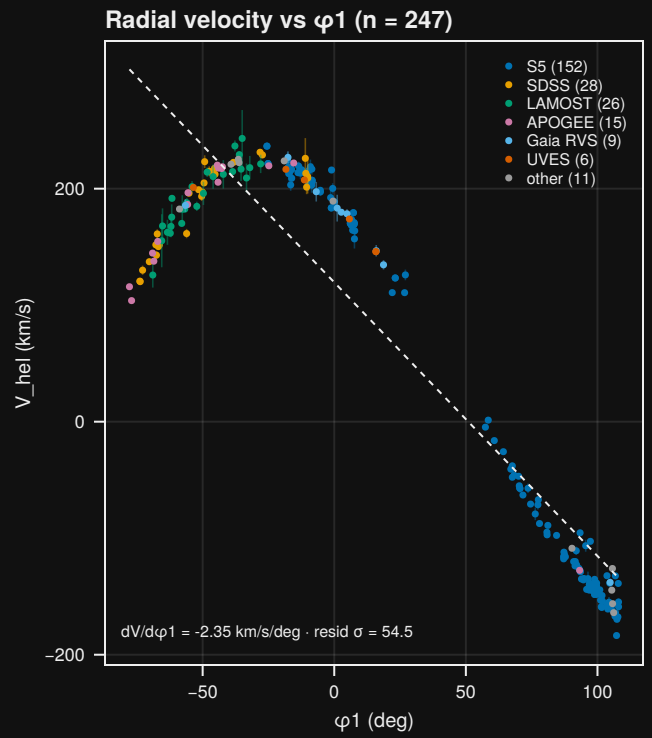
Kinematics · proper motion · parallax · radial velocity

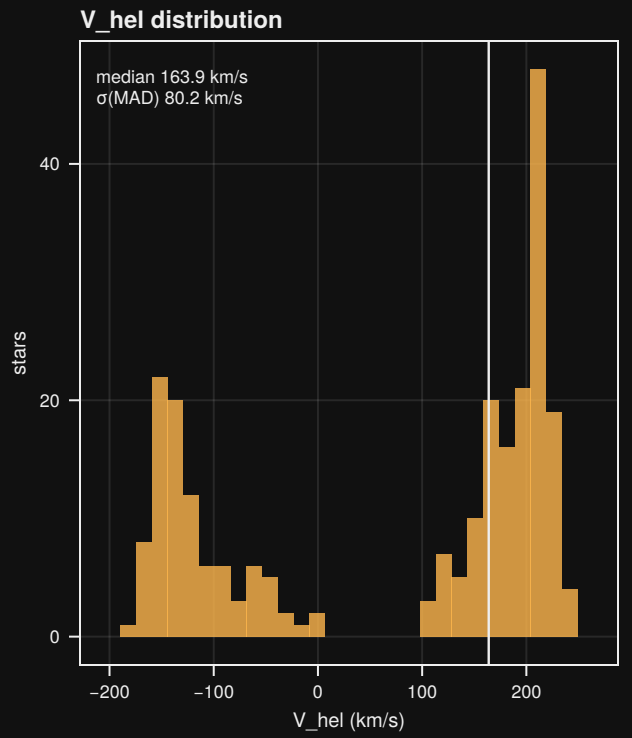






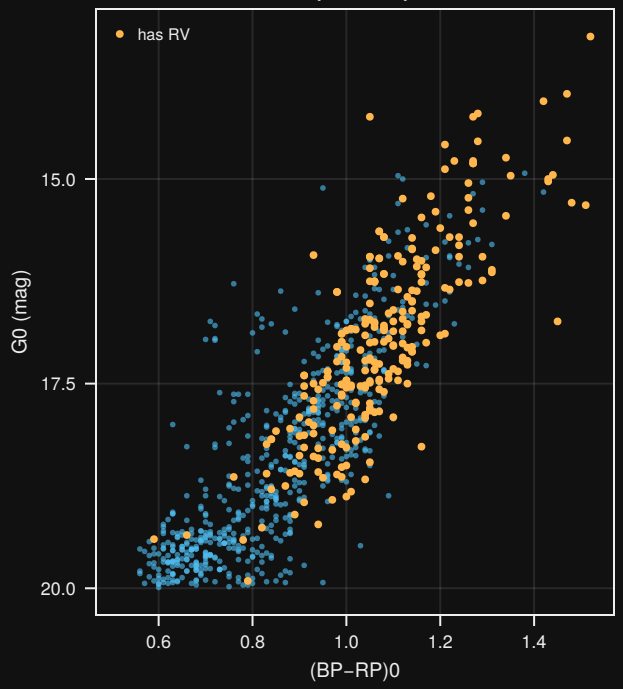




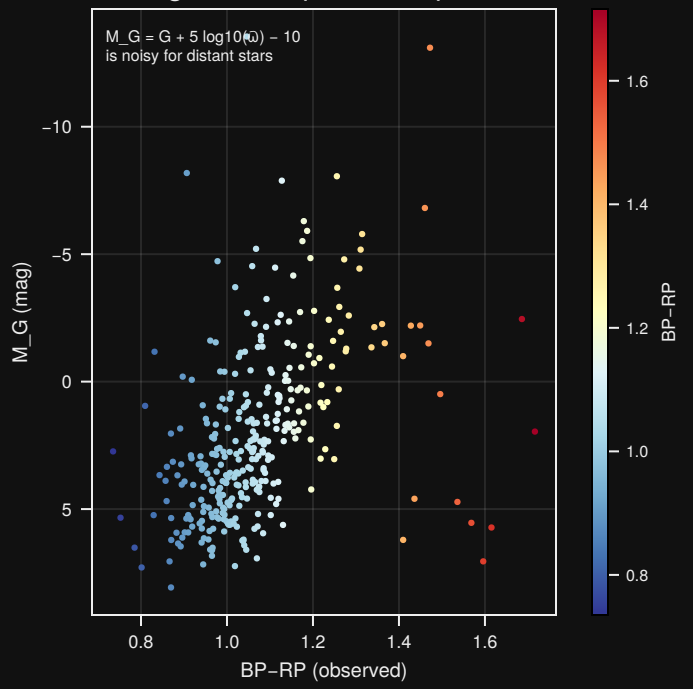


Orphan

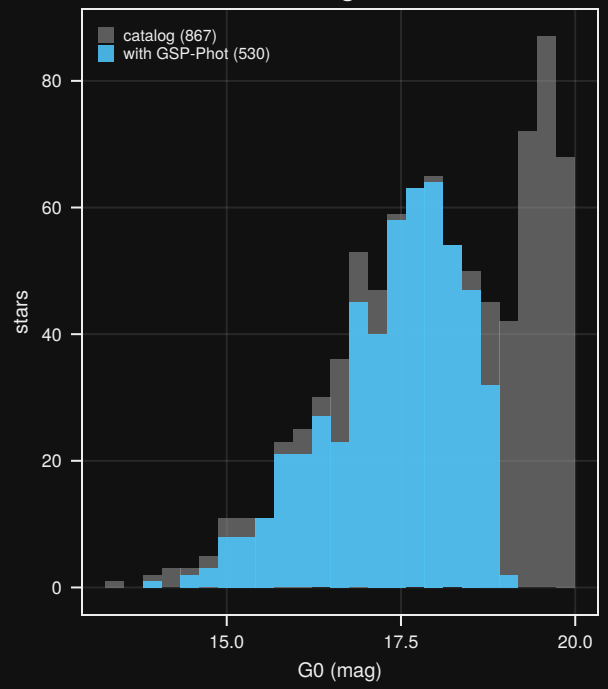
CMD · dereddened (n = 867)



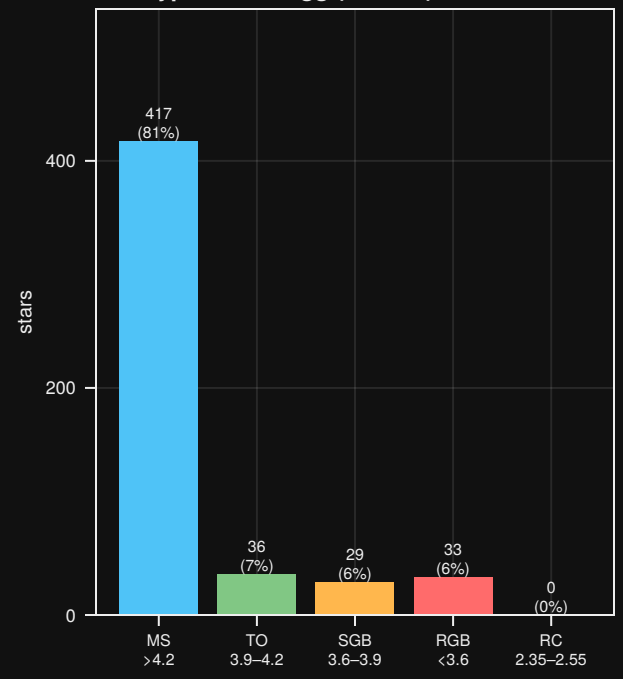
HR diagram · ϖ>0 (n = 331/515)



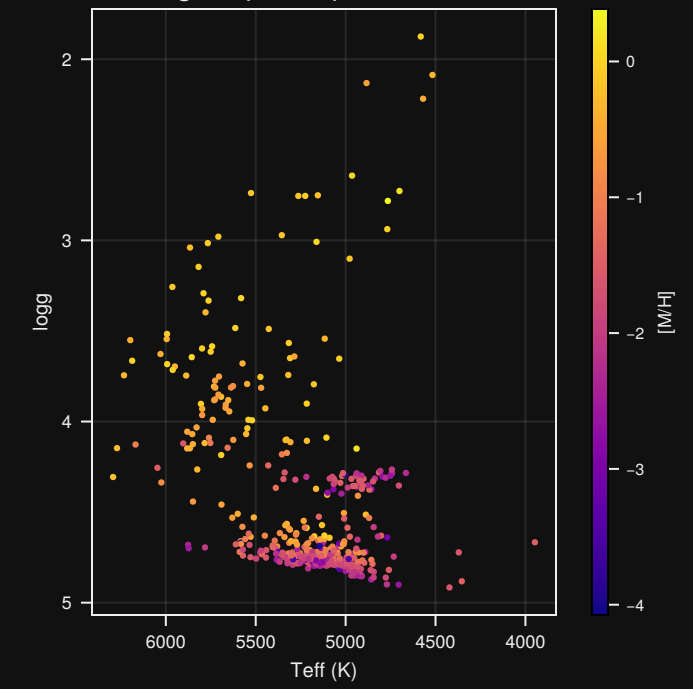
G0 & GSP-Phot coverage



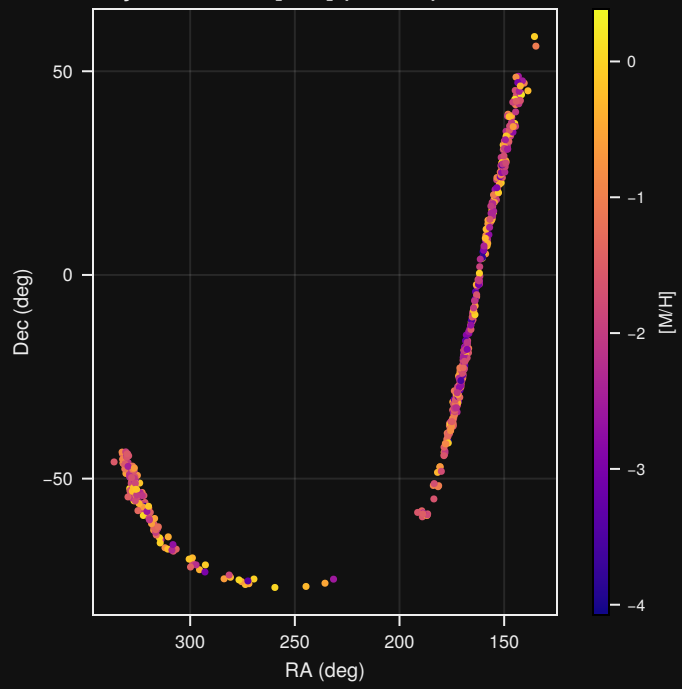
Star types from logg (n = 515)



Kiel diagram (n = 515)



Sky · GSP-Phot [M/H] (n = 515)

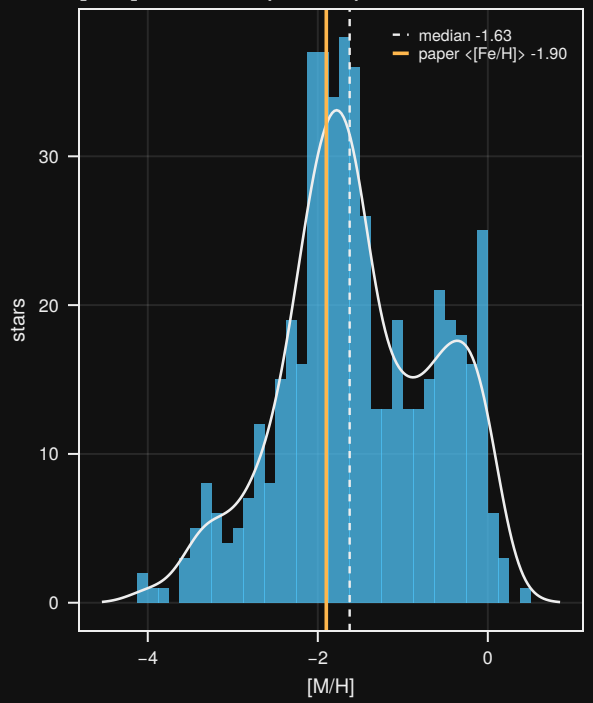


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

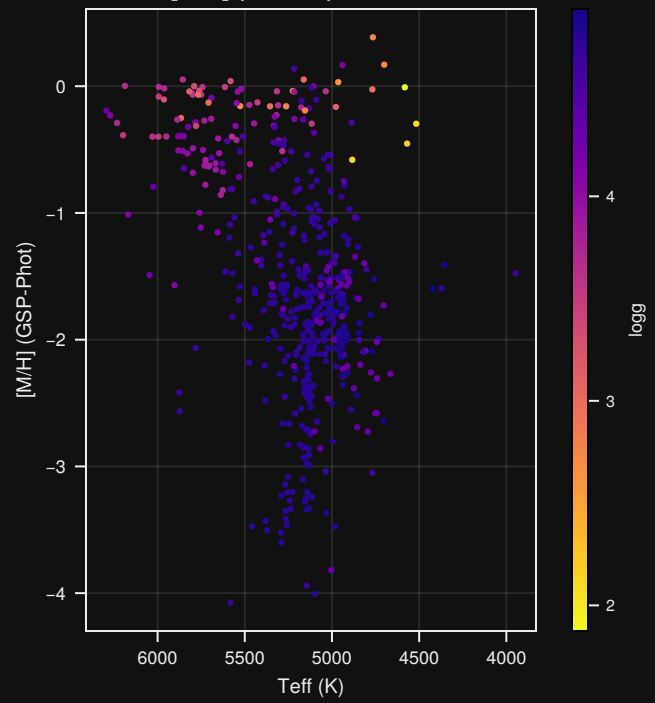
Orphan

Metallicity · age · astrometric quality

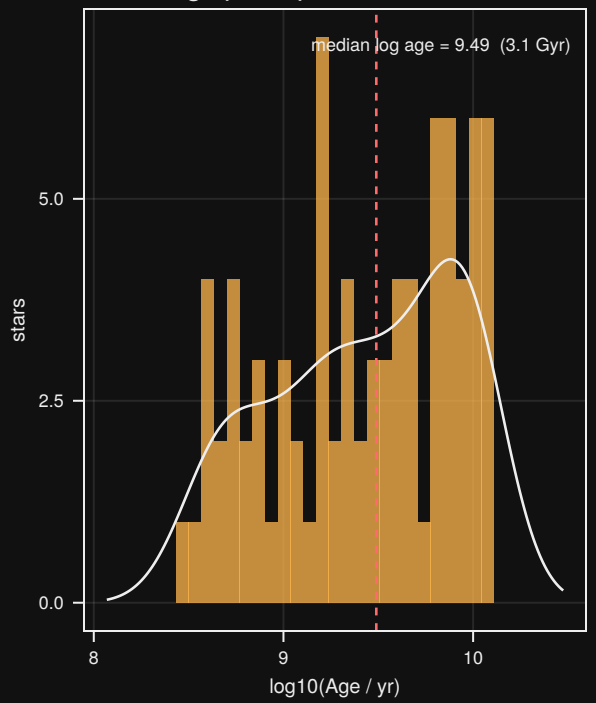
[M/H] GSP-Phot (n = 515)



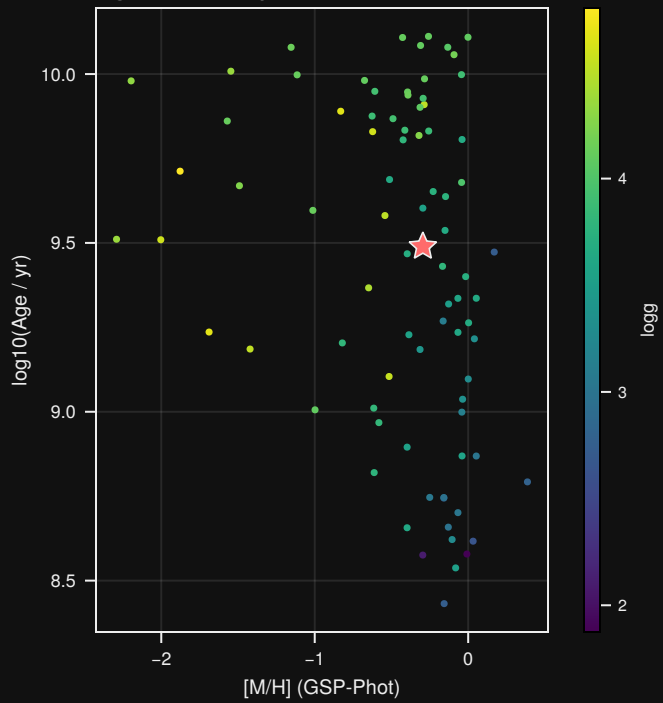
Teff vs [M/H] (n = 515)



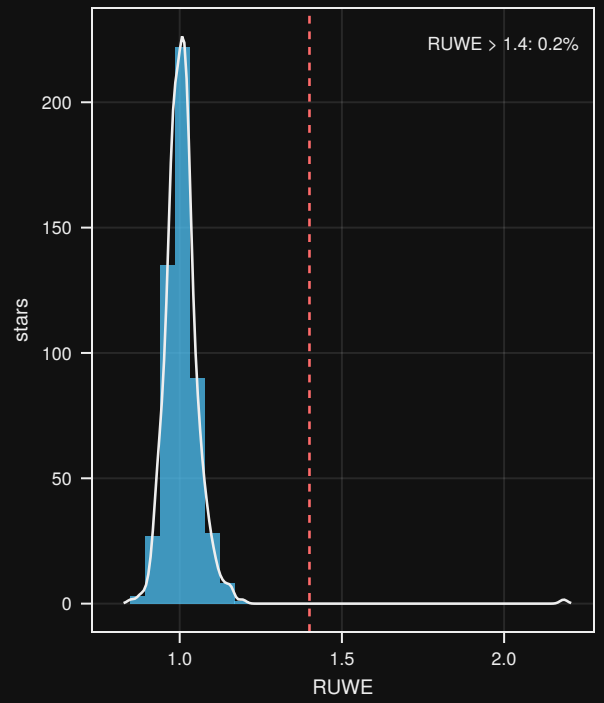
FLAME age (n = 83)



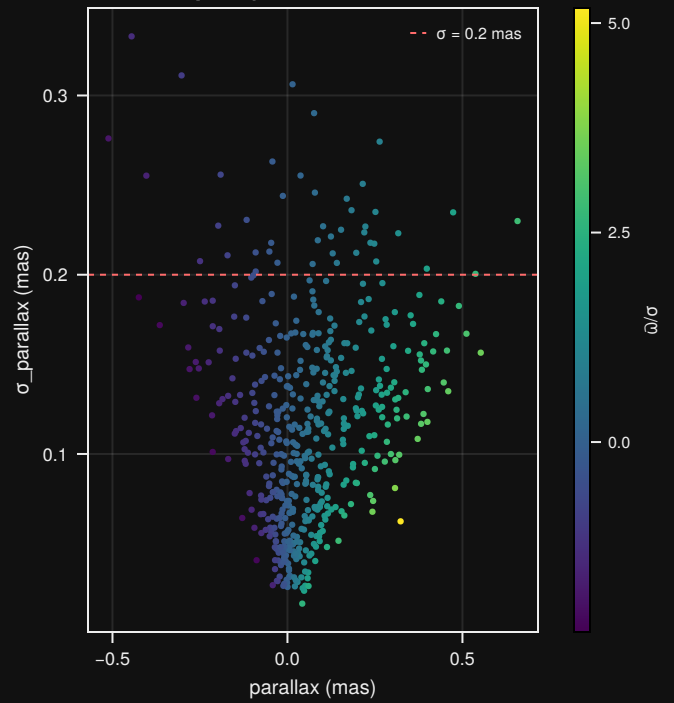
Age–metallicity (n = 83)



RUWE (n = 515)

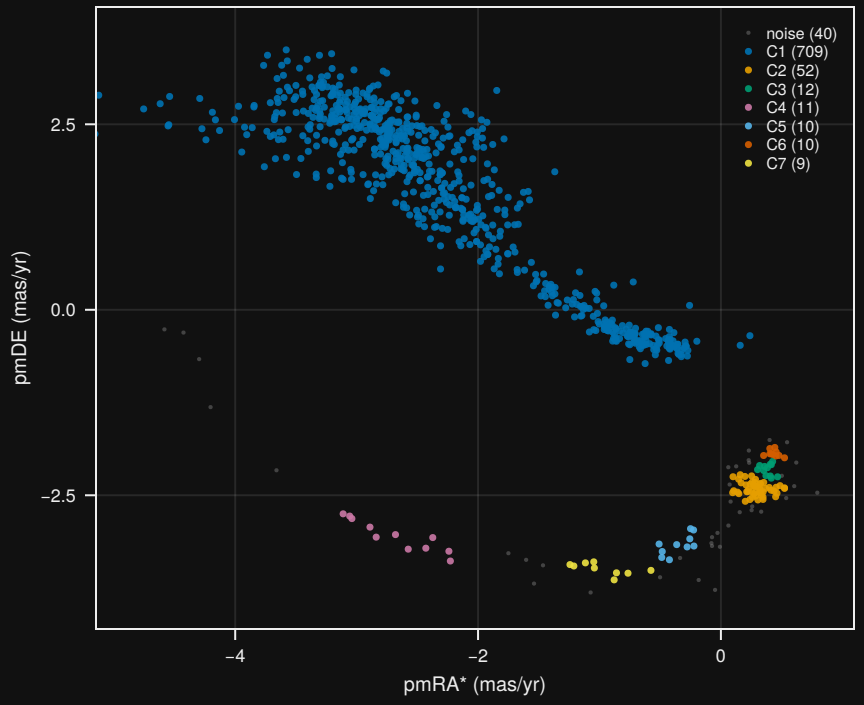


Parallax quality (n = 515)

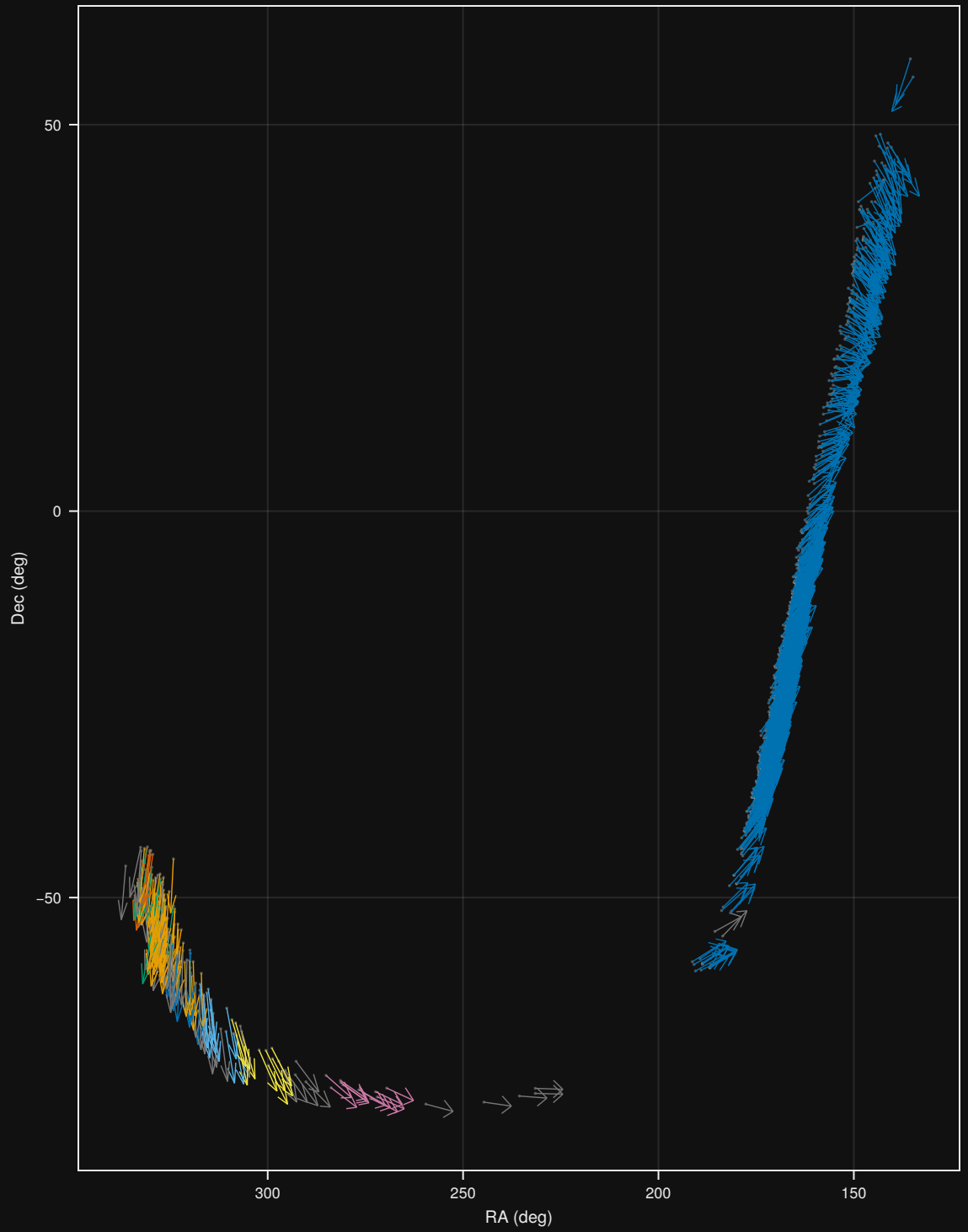


Orphan

PM clustering · HDBSCAN · 9 cluster(s) · noise 5%



Proper-motion directions (arrows) coloured by cluster



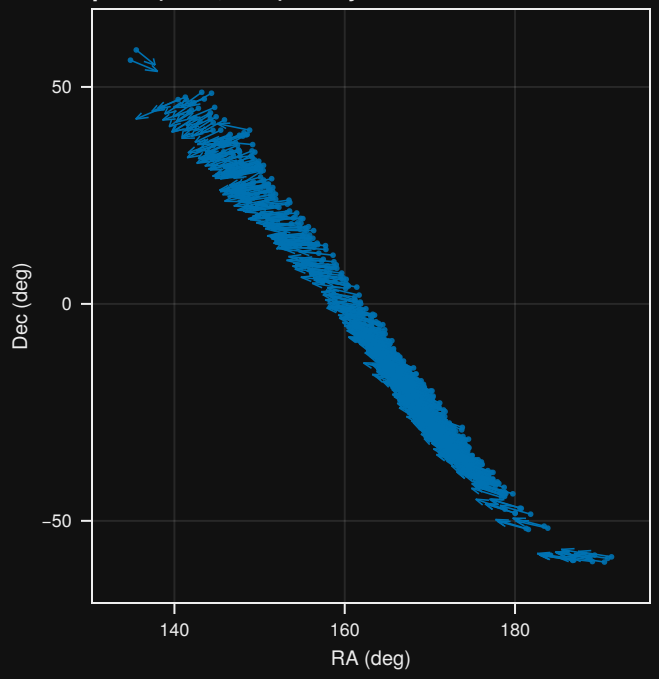
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	40	-0.02	-2.56	0.57	-1.52	19
C1	709	-2.49	2.00	0.87	-1.69	153
C2	52	0.29	-2.42	0.09	-1.56	31
C3	12	0.40	-2.14	0.04	-1.46	8
C4	11	-2.68	-3.06	0.46	-0.40	10
C5	10	-0.32	-3.17	0.14	-1.31	4
C6	10	0.44	-1.95	0.04	-1.62	7
C7	9	-1.04	-3.48	0.25	-1.45	6
C8	9	0.09	-2.82	0.11	-1.31	7

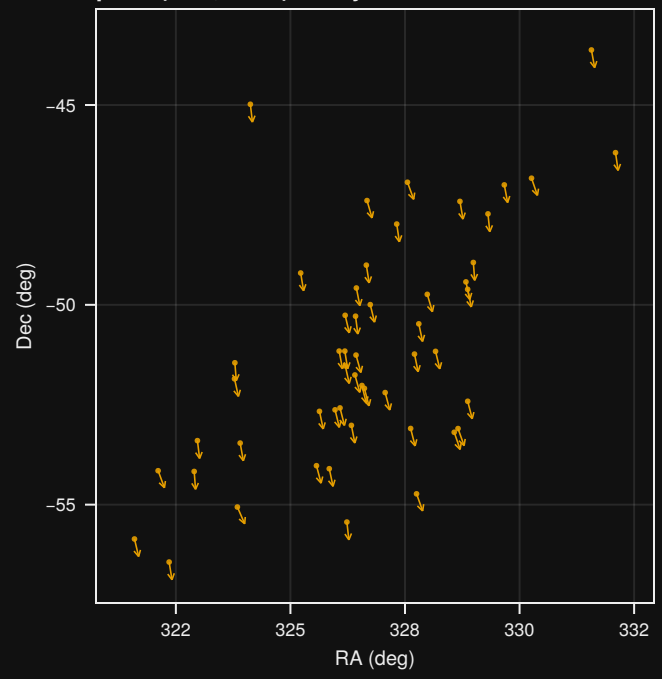
Orphan

Per-cluster proper-motion vector maps

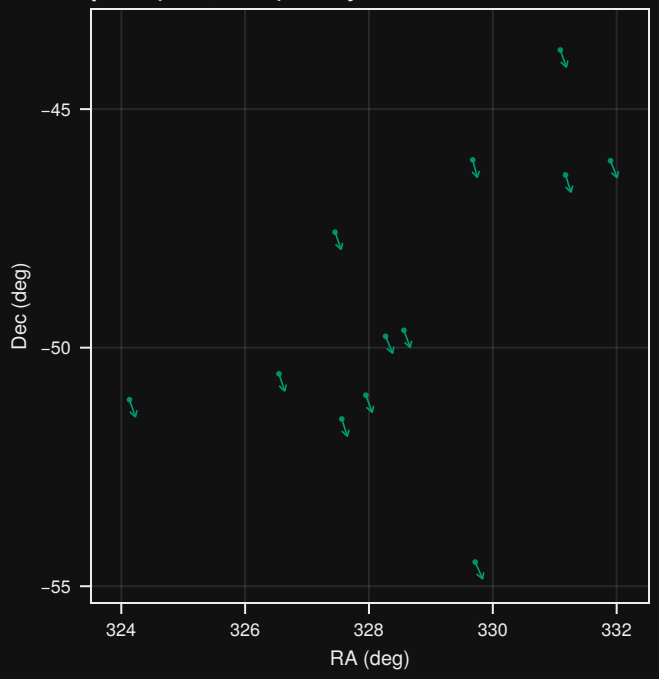
Cluster C1 · n = 709
pm = (-2.49, 2.00) mas/yr



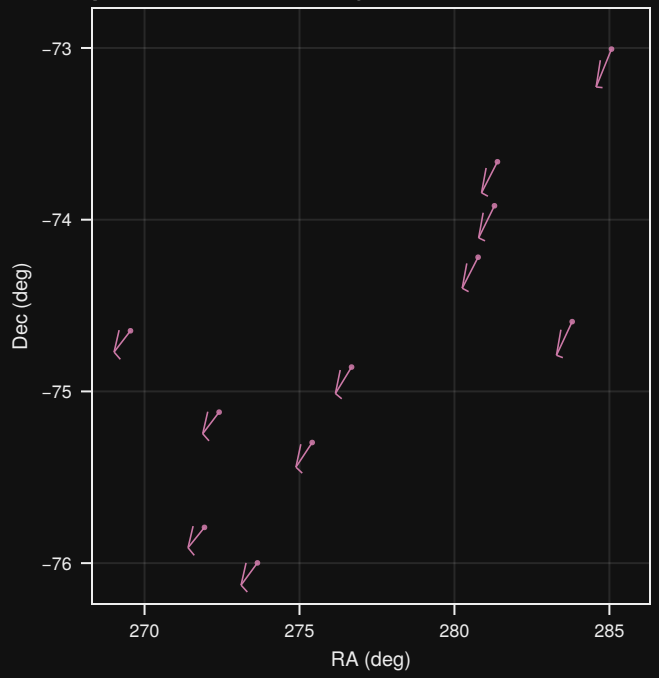
Cluster C2 · n = 52
pm = (0.29, -2.42) mas/yr



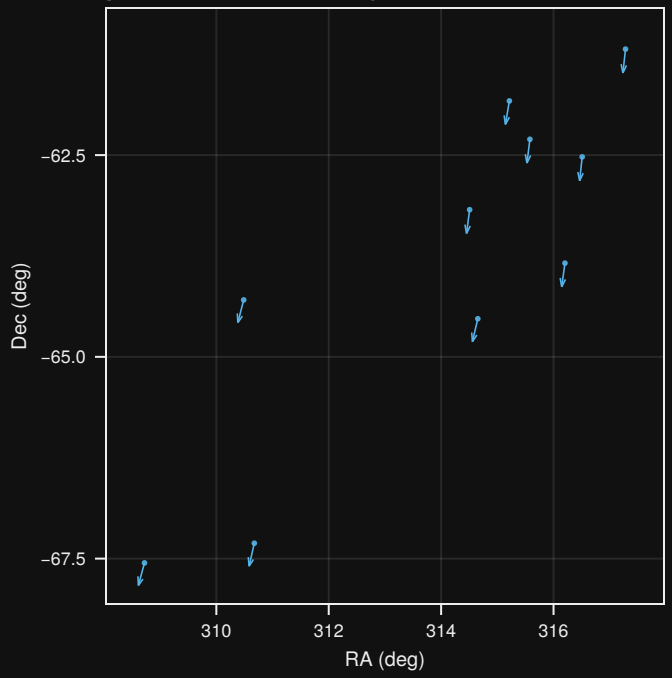
Cluster C3 · n = 12
pm = (0.40, -2.14) mas/yr



Cluster C4 · n = 11
pm = (-2.68, -3.06) mas/yr



Cluster C5 · n = 10
pm = (-0.32, -3.17) mas/yr



Noise · n = 40

